

FOLoRA: Fisher-Orthogonalized Low-Rank Adaptation for Continual Learning

Jiang Tao^{a,*}

^a*School of Software Technology, Zhejiang University, Ningbo, China*

Abstract

Parameter-efficient fine-tuning (PEFT) methods such as low-rank adaptation (LoRA) freeze a pre-trained backbone and learn only a small number of trainable parameters, making them attractive for continual learning on resource-constrained devices. However, even with few trainable parameters, sequentially fine-tuning a shared LoRA adapter across tasks still suffers from catastrophic forgetting, because the low-rank subspace is repeatedly overwritten by new tasks. In this paper we propose **FOLoRA** (Fisher-Orthogonalized Low-Rank Adaptation), a replay-free, fixed-budget method that mitigates forgetting by protecting the directions that past tasks deem important. FOLoRA maintains, for each adapted layer, an accumulated second-order importance kernel estimated from the empirical Fisher information restricted to the adapter parameters. Before learning a new task, the top eigendirections of this kernel form a *importance-weighted protected subspace*; during training, a regularizer penalizes the alignment of the adapter displacement with these directions, weighted by their eigenvalues. This yields a principled stability-plasticity trade-off: strongly important directions are preserved, while unimportant directions remain available for reuse. We further derive a forgetting upper bound that decomposes into Fisher-weighted subspace overlaps and show that FOLoRA’s regularizer directly minimizes its dominant term. Experiments on Split CIFAR-100 and Split ImageNet-R show that FOLoRA consistently outperforms sequential LoRA, EWC-LoRA, O-LoRA, L2P and CODA-Prompt in both average accuracy and forgetting, with no replay buffer and no growth in the number of parameters.

*Corresponding author

Email address: jiangtao@example.com (Jiang Tao)

Keywords: continual learning, catastrophic forgetting, parameter-efficient fine-tuning, low-rank adaptation, Fisher information

1. Introduction

Continual learning aims to learn a sequence of tasks without catastrophically forgetting previously acquired knowledge [1]. The dominant strategies — rehearsal [2], regularization [3, 4], and dynamic architectures [5] — were largely developed in the regime of training a model from scratch. Recently, however, the community has shifted toward *pre-trained models* (e.g., Vision Transformers) combined with *parameter-efficient fine-tuning* (PEFT), in which the backbone is frozen and only a tiny set of parameters is learned per task [6, 7, 8].

Low-rank adaptation (LoRA) [9] is a particularly attractive PEFT primitive for continual learning: a shared LoRA adapter keeps the total parameter count fixed regardless of the number of tasks, which is ideal for resource-constrained deployment. However, sequentially fine-tuning a shared adapter still suffers from catastrophic forgetting — although the number of trainable parameters is small, the low-rank subspace is repeatedly overwritten, and the representational structure learned for earlier tasks is destroyed.

A line of recent work addresses this by allocating *orthogonal* low-rank subspaces to different tasks [8, 10]. These methods prevent interference by making the new task’s adapter (nearly) orthogonal to those of previous tasks. While effective, they exhibit two limitations. First, they protect all directions of the past subspace *equally*, ignoring the fact that different directions carry vastly different amounts of information about a task. Second, most of them allocate a *new* adapter per task, so the parameter count grows linearly with the number of tasks.

In this paper we propose **FOLoRA** (Fisher-Orthogonalized Low-Rank Adaptation), a replay-free, fixed-budget method that mitigates forgetting by protecting past tasks’ *important* directions in proportion to their importance. Our contributions are:

1. We frame forgetting in low-rank continual learning through a second-order lens and derive an upper bound that decomposes forgetting into Fisher-weighted subspace overlaps (Sec. 4).
2. We propose FOLoRA, which estimates a per-layer second-order importance kernel from the empirical Fisher information restricted to the

adapter parameters, forms an importance-weighted protected subspace from its top eigendirections, and regularizes the adapter to stay orthogonal to these directions (Sec. 3).

3. We show that FOLoRA’s regularizer is exactly the Lagrangian relaxation of the derived bound, realizing a principled stability–plasticity trade-off, and that it subsumes equal-weight orthogonalization (O-LoRA) as a special case.
4. On Split CIFAR-100 and Split ImageNet-R, FOLoRA consistently outperforms sequential LoRA, EWC-LoRA, O-LoRA, L2P and CODA-Prompt in both average accuracy and forgetting, with no replay buffer and no growth in parameters (Sec. 5).

2. Related Work

Continual learning. Classical continual learning methods fall into three families. Rehearsal methods store and replay a subset of past data [2]; regularization methods add a penalty that slows down learning on parameters important to past tasks, most notably EWC via the diagonal Fisher information [3] and SI/MAS via synaptic importance [4, 11]; dynamic methods grow the network per task [5, 12]. Our method is a regularization method, but it operates on the low-rank subspace of a frozen backbone rather than on raw parameters.

Parameter-efficient continual learning with pre-trained models. A recent and effective line of work keeps a pre-trained backbone frozen and learns only prompts or adapters. Prompt-based methods — L2P [6], DualPrompt, and CODA-Prompt [7] — learn a pool of prompts and select them per input. LoRA-based methods attach low-rank adapters. O-LoRA [8] assigns orthogonal subspaces to successive tasks by orthogonally initializing each new adapter; InfLoRA [10] further aligns the subspaces with the input feature distribution of the pre-trained weights. FOLoRA differs in two ways: it uses a single fixed-budget adapter rather than one adapter per task, and it weights the protected directions by their Fisher importance rather than treating them uniformly.

Second-order methods and gradient projection. EWC [3] uses the diagonal Fisher as a per-parameter importance weight. Gradient-projection methods — GPM [13] and its variants — maintain a subspace of past gradients and

project new gradients onto its null space. FOLoRA combines ideas from both: it uses full second-order structure restricted to the adapter parameters, and applies it as a soft, importance-weighted orthogonal projection rather than a hard null-space constraint.

Low-rank adaptation.. LoRA [9] approximates weight updates by low-rank matrices and has become a standard PEFT primitive. Subsequent work studies initialization, rank allocation, and merging. We build directly on LoRA and extend it with a second-order anti-forgetting mechanism specialized for continual learning.

3. Method

3.1. Preliminaries

Low-rank adaptation (LoRA). LoRA [9] freezes a pre-trained weight matrix $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and injects a trainable low-rank residual,

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA, \quad (1)$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$ and $r \ll \min(d_{\text{in}}, d_{\text{out}})$. The scalar α/r absorbs the rank normalization. The forward pass of an adapted layer is

$$h = W_0 x + \frac{\alpha}{r} BAx = W_0 x + \delta, \quad \delta := \frac{\alpha}{r} BAx \in \mathbb{R}^{d_{\text{out}}}, \quad (2)$$

where δ denotes the adapter’s residual contribution.

Class-incremental continual learning (CIL). We consider a sequence of T tasks $\{\mathcal{T}_t\}_{t=1}^T$. Task $\mathcal{T}_t$ provides a training set $\mathcal{D}_t = \{(x_i, y_i)\}$ over k new classes, and the classifier head is grown to cover all classes seen so far. At test time the task identity is not available, so prediction is made over the union of all seen classes. A single shared LoRA adapter is trained sequentially; its parameter count is $2r(d_{\text{in}} + d_{\text{out}})$ per layer and is *independent* of T .

3.2. Motivation: forgetting as subspace overlap

Let ΔW^τ denote the adapter after task $\mathcal{T}_\tau$ and let θ_τ collect all trainable parameters at that point. When a later task displaces the adapter by $\delta W = \Delta W - \Delta W^\tau$, a second-order Taylor expansion of task τ ’s loss around θ_τ gives

$$\mathcal{L}_\tau(\theta_\tau + \delta W) \approx \mathcal{L}_\tau(\theta_\tau) + \langle \nabla_{\Delta W} \mathcal{L}_\tau, \delta W \rangle + \frac{1}{2} \delta W^\top F_\tau \delta W, \quad (3)$$

where $F_\tau = \mathbb{E}_{x \sim \mathcal{D}_\tau} [\nabla \log p(y|x) \nabla \log p(y|x)^\top]$ is the empirical Fisher information matrix of task τ . At convergence $\nabla_{\Delta W} \mathcal{L}_\tau \approx 0$, hence the *forgetting* of task τ — the increase in its loss caused by subsequent learning — is dominated by the quadratic term $\frac{1}{2} \delta W^\top F_\tau \delta W$. Forgetting is therefore driven by the alignment of the new displacement δW with the directions in which task τ 's loss is sharply curved (i.e., the top eigendirections of F_τ).

This observation separates FOLoRA from prior work: O-LoRA [8] treats every direction of the past low-rank subspace as equally important, whereas Eq. (3) shows that directions should be protected *in proportion to their curvature*. Protecting all directions equally over-constrains the adapter and wastes the fixed rank budget on directions that carry little information; protecting directions by their Fisher weight realizes the optimal stability–plasticity trade-off.

3.3. FOLoRA

We now describe FOLoRA. For each adapted layer, let $\theta \in \mathbb{R}^d$ denote the flattened adapter parameters (the LoRA matrices A and B , so $d = r(d_{\text{in}} + d_{\text{out}})$), and let $g_\theta = \nabla_\theta \log p(y|x)$ denote the gradient of the log-likelihood with respect to these parameters.

Importance estimation.. After finishing task $\mathcal{T}_\tau$, we estimate a per-layer *importance kernel* from the per-sample parameter-gradient autocorrelation,

$$F_\tau = \mathbb{E}_{x \sim \mathcal{D}_\tau} [g_\theta(x) g_\theta(x)^\top] \in \mathbb{R}^{d \times d}, \quad (4)$$

which is the empirical Fisher information of task τ restricted to the adapter parameters. To keep the estimate tractable we never materialize the full $d \times d$ matrix; we collect the per-sample gradients into a matrix (one row per sample) and retain only its top- k singular directions, i.e., a rank- k approximation of F_τ . We accumulate these kernels across tasks,

$$\bar{F} = \sum_{\tau < t} F_\tau. \quad (5)$$

Protected subspace.. At the start of task $\mathcal{T}_t$, we eigendecompose the accumulated kernel $\bar{F} = \sum_j \sigma_j u_j u_j^\top$, where $\{\sigma_j\}_j$ are the eigenvalues sorted in decreasing order and $\{u_j\}$ the corresponding orthonormal eigenvectors in $\mathbb{R}^d$. We retain the top- k eigendirections as an *importance-weighted protected subspace* $\mathcal{P}_t = \{(\sigma_j, u_j)\}_{j=1}^k$.

Orthogonalization regularizer. Let θ^{t-1} denote the adapter parameter snapshot recorded after task $\mathcal{T}_{t-1}$, and let $\delta\theta = \theta - \theta^{t-1}$ be the parameter displacement accumulated while training task $\mathcal{T}_t$. During the training of task $\mathcal{T}_t$, FOLoRA minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\theta) + \lambda \underbrace{\sum_{j=1}^k \sigma_j \|u_j^\top \delta\theta\|_2^2}_{\Omega(\delta\theta)}, \quad (6)$$

where $\lambda > 0$ controls the strength. The term $\|u_j^\top \delta\theta\|_2^2$ penalizes the alignment of the *displacement* $\delta\theta = \theta - \theta^{t-1}$ —rather than the absolute θ —with the protected direction u_j , weighted by the importance σ_j . Using the displacement is essential: at the start of task $\mathcal{T}_t$ we have $\delta\theta = 0$, so the regularizer vanishes and the adapter’s existing alignment with past tasks is preserved; only *new* movement into protected directions is penalized. Penalizing the absolute θ would instead push the adapter away from the past tasks’ directions and actively erase them, defeating the purpose of protection. Since the regularizer is a differentiable function of A and B , it integrates with standard backpropagation and requires no hard projection, no replay buffer, and no task identity at inference.

Why importance weighting matters. Setting all $\sigma_j = 1$ reduces Ω to an equal-weight orthogonalization akin to the spirit of O-LoRA. In contrast, FOLoRA’s eigenvalue weighting (i) strongly suppresses reuse of high-curvature directions, which is where forgetting is concentrated, and (ii) leaves low-curvature directions nearly free, so the adapter can reuse them for new tasks without wasting rank. This is the key mechanism behind the stability–plasticity trade-off quantified in Sec. 4.

The full algorithm is summarized in Algorithm 1.

3.4. Relation to prior methods

- **EWC** [3] penalizes per-parameter quadratic displacement $\sum_i F_{ii}(\theta_i - \theta_i^*)^2$ using the *diagonal* Fisher. FOLoRA instead uses the *full* second-order structure restricted to the adapter parameters and applies it as a rank- k subspace projection rather than a pointwise penalty, which captures the parameter correlations that the diagonal approximation discards and is more faithful to the low-rank geometry.

Algorithm 1 FOLoRA for class-incremental continual learning

- 1: **Input:** frozen backbone; LoRA rank r ; strength λ ; budget k .
 - 2: Initialize shared LoRA adapters $\Delta W^{(l)} = \frac{\alpha}{r} B^{(l)} A^{(l)}$ for each adapted layer l .
 - 3: $\bar{F}^{(l)} \leftarrow \mathbf{0}$ for all l .
 - 4: **for** each task $\mathcal{T}_t$, $t = 1, \dots, T$ **do**
 - 5: **if** $t > 1$: eigendecompose $\bar{F}^{(l)} = \sum_j \sigma_j u_j u_j^\top$, keep top- k $\{(\sigma_j, u_j)\}$.
 - 6: Train adapter parameters $\theta^{(l)}$ and the grown classifier head by minimizing $\mathcal{L}_{\text{CE}} + \lambda \sum_l \sum_j \sigma_j \|u_j^\top (\theta^{(l)} - \theta_{t-1}^{(l)})\|_2^2$.
 - 7: Record adapter snapshot $\theta_{t-1}^{(l)} \leftarrow \theta^{(l)}$.
 - 8: Estimate $F_t^{(l)} = \mathbb{E}[g_\theta g_\theta^\top]$ on $\mathcal{D}_t$; update $\bar{F}^{(l)} \leftarrow \bar{F}^{(l)} + F_t^{(l)}$.
 - 9: **end for**
 - 10: **Output:** a single shared adapter with fixed parameter budget.
-

- **O-LoRA** [8] allocates a new adapter per task and initializes it orthogonally to previous tasks' input subspaces, treating all directions equally; FOLoRA keeps a *single* fixed-budget adapter and protects directions in proportion to their Fisher importance.
- **GPM** [13] and related null-space methods project gradients onto a null space; FOLoRA realizes a *soft*, importance-weighted orthogonal projection that avoids the rank saturation of hard projections.

4. Theoretical Analysis

We derive an upper bound on the forgetting of previously learned tasks under sequential low-rank adaptation, and show that the FOLoRA regularizer minimizes its dominant term.

4.1. Setup

Let θ_τ denote the trainable parameters (adapter plus classifier) after learning task $\mathcal{T}_\tau$, and let the adapter component be ΔW^τ . For a later task $\mathcal{T}_t$ ($t > \tau$), let $\delta W = \Delta W^t - \Delta W^\tau$ be the adapter displacement. We measure the forgetting of task τ by the increase of its empirical loss,

$$\Delta \mathcal{L}_\tau = \mathcal{L}_\tau(\theta_t) - \mathcal{L}_\tau(\theta_\tau). \quad (7)$$

4.2. A second-order forgetting bound

We make the standard assumption (used by EWC [3] and GPM [13]) that each task is trained to a (near-)stationary point, so $\nabla_{\theta} \mathcal{L}_{\tau}(\theta_{\tau}) \approx 0$.

Theorem 1 (Forgetting bound). *Suppose the empirical loss $\mathcal{L}_{\tau}$ is twice differentiable and θ_{τ} is a stationary point of $\mathcal{L}_{\tau}$. Then, up to second order,*

$$\Delta \mathcal{L}_{\tau} \leq \frac{1}{2} \delta W^{\top} F_{\tau} \delta W + \mathcal{O}(\|\delta W\|^3), \quad (8)$$

where $F_{\tau} = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\tau}} [\nabla \log p_{\theta}(y|x) \nabla \log p_{\theta}(y|x)^{\top}]$ is the empirical Fisher of task τ restricted to the adapter parameters.

Proof. Taylor-expanding $\mathcal{L}_{\tau}$ at θ_{τ} along the displacement $\delta\theta = \theta_t - \theta_{\tau}$ gives

$$\mathcal{L}_{\tau}(\theta_t) = \mathcal{L}_{\tau}(\theta_{\tau}) + \langle \nabla \mathcal{L}_{\tau}(\theta_{\tau}), \delta\theta \rangle + \frac{1}{2} \delta\theta^{\top} H_{\tau} \delta\theta + \mathcal{O}(\|\delta\theta\|^3), \quad (9)$$

where H_{τ} is the Hessian. Since $\nabla \mathcal{L}_{\tau}(\theta_{\tau}) \approx 0$, the linear term vanishes. Under the Fisher-information approximation $H_{\tau} \approx F_{\tau}$ valid for (negative log-)likelihood objectives near the optimum [14], and restricting the displacement to the adapter parameters δW , we obtain Eq. (8). $\square$

Because the classifier head is shared and small, the dominant source of forgetting is the adapter displacement; Eq. (8) isolates exactly this term.

4.3. FOLoRA minimizes the bound

Summing Eq. (8) over all previous tasks $\tau < t$ yields

$$\sum_{\tau < t} \Delta \mathcal{L}_{\tau} \lesssim \frac{1}{2} \delta W^{\top} \left(\sum_{\tau < t} F_{\tau} \right) \delta W = \frac{1}{2} \delta W^{\top} \bar{F} \delta W, \quad (10)$$

where we identify the accumulated Fisher $\sum_{\tau < t} F_{\tau}$ with the accumulated importance kernel $\bar{F}$ of Eq. (5) (restricted to the adapter parameters). Using the eigendecomposition $\bar{F} = \sum_j \sigma_j u_j u_j^{\top}$,

$$\sum_{\tau < t} \Delta \mathcal{L}_{\tau} \lesssim \frac{1}{2} \sum_j \sigma_j (u_j^{\top} \delta W)^2. \quad (11)$$

Equation (11) shows that forgetting is bounded by a sum of terms $\sigma_j (u_j^{\top} \delta W)^2$, i.e., the squared projection of the displacement onto each eigendirection u_j , weighted by its importance σ_j . The FOLoRA regularizer in Eq. (6),

$$\Omega(\delta W) = \sum_j \sigma_j \|u_j^{\top} \delta W\|_F^2, \quad \delta W = \Delta W - \Delta W^{t-1}, \quad (12)$$

is precisely the Lagrangian relaxation of the constraint that the adapter displacement remain orthogonal to the high-importance eigendirections: minimizing Ω drives $\|u_j^\top \delta W\|$ toward zero for the largest σ_j , which in turn minimizes the dominant terms of Eq. (11). We therefore interpret FOLoRA as the algorithm that *directly optimizes an upper bound on catastrophic forgetting*, with λ controlling the stability–plasticity trade-off and k controlling the rank of the protected subspace. Following the standard single-reference approximation used by EWC [3], the algorithm stores one adapter snapshot ΔW^{t-1} (after the previous task) and uses it as the reference for all past tasks; the per-task references ΔW^τ in Eq. (10) are replaced by ΔW^{t-1} .

4.4. Connection to equal-weight orthogonalization

If all eigenvalues are set to $\sigma_j = 1$, the bound in Eq. (11) becomes $\frac{1}{2}\|\bar{U}^\top \delta W\|^2$ with an equal-weight orthonormal basis $\bar{U}$, which is the geometry underlying O-LoRA-style orthogonal subspace methods. FOLoRA’s eigenvalue weighting is strictly more expressive: it can concentrate protection on the few high-curvature directions and thereby achieve a smaller forgetting for the same rank budget, or equivalently the same forgetting with a smaller protected rank k . This prediction is verified empirically by the ablation over λ and k in Sec. 5.3.

5. Experiments

5.1. Setup

We use a ViT-B/16 backbone pre-trained on ImageNet-1k, frozen throughout, with LoRA adapters (rank $r = 16$, $\alpha = 16$) attached to the MLP down-projection of every transformer block (12 adapters in total). The classifier head is a single linear layer grown as new classes arrive. All methods share the identical backbone and head; they differ only in the adapter regularization.

Datasets. Split CIFAR-100 (20 tasks $\times$ 5 classes) and Split ImageNet-R (20 tasks $\times$ 10 classes), both under the class-incremental protocol (task identity unavailable at test time).

Baselines. (i) *Seq-LoRA*: sequential fine-tuning of the shared adapter with no protection (lower bound); (ii) *EWC-LoRA*: diagonal Fisher on the LoRA parameters [3]; (iii) *O-LoRA*: per-task adapters with orthogonal initialization [8]; (iv) *L2P*: a shared pool of prompts selected per input by key–query matching [6]; (v) *CODA-Prompt*: a set of prompt components combined by attention [7]; (vi) *FOLoRA* (ours).

Table 1: Class-incremental results (mean $\pm$ std). Adapter-based methods are trained for 5 epochs (10 seeds on CIFAR-100, 5 on ImageNet-R); prompt-based methods for 20 epochs (3 seeds). $\overline{\text{ACC}}$: average accuracy; FGT: average forgetting. Lower FGT is better. Best in **bold**.

Method	Split CIFAR-100		Split ImageNet-R	
	$\overline{\text{ACC}}$	FGT	$\overline{\text{ACC}}$	FGT
Seq-LoRA	7.19 $\pm$ 0.93	90.90 $\pm$ 1.05	6.55 $\pm$ 0.36	84.42 $\pm$ 0.26
EWC-LoRA	9.17 $\pm$ 0.89	88.76 $\pm$ 1.00	8.17 $\pm$ 0.75	82.92 $\pm$ 0.95
O-LoRA	7.74 $\pm$ 0.79	90.01 $\pm$ 0.96	6.38 $\pm$ 0.43	83.44 $\pm$ 0.27
L2P	8.67 $\pm$ 0.55	89.08 $\pm$ 0.58	7.05 $\pm$ 0.32	80.22 $\pm$ 0.36
CODA-Prompt	9.35 $\pm$ 0.52	88.52 $\pm$ 0.65	6.93 $\pm$ 0.27	79.54 $\pm$ 0.59
FOLoRA (ours)	10.07$\pm$1.54	87.65$\pm$1.43	9.56$\pm$0.83	79.44$\pm$0.62

Training. AdamW with learning rate 10^{-3} , batch size 32, and bf16 mixed precision. We train the adapter-based methods (Seq-LoRA, EWC-LoRA, O-LoRA, FOLoRA) for 5 epochs per task, which avoids over-fitting and keeps the empirical Fisher meaningful (Sec. 5.4); the prompt-based methods (L2P, CODA-Prompt) are trained for 20 epochs per task, their optimal setting since their key-based prompt selection benefits from more epochs. For FOLoRA we select λ and k by a small grid search over $\lambda \in \{10, 30, 100, 300, 1000\}$ and $k \in \{16, 64\}$, taking $\lambda = 300$, $k = 16$ as the default; the importance kernel is estimated from up to 300 per-sample gradients per task. EWC’s regularization strength is likewise tuned on CIFAR-100, reaching its optimum at $\lambda = 30$. We report the mean and standard deviation over 10 seeds on CIFAR-100 (5 seeds on ImageNet-R) for the adapter-based methods, and 3 seeds for the prompt-based methods.

Metrics. Average accuracy over all tasks after the final task ($\overline{\text{ACC}}$), average forgetting (FGT), and average incremental accuracy.

5.2. Main results

On both benchmarks FOLoRA attains the highest average accuracy and the lowest average forgetting, including over the prompt-based methods at their optimal epoch budget. The improvement is largest on ImageNet-R, where FOLoRA improves accuracy by +1.39 points over the strongest baseline (EWC-LoRA) while reducing forgetting by 3.48 points; on CIFAR-100 the gains over EWC-LoRA are +0.90 and -1.11 points, respectively, and

Table 2: Ablation of λ (with $k = 16$) and k (with $\lambda = 10$) on Split CIFAR-100, seed 0.

λ	$\overline{\text{ACC}}$	FGT
10	9.20	89.22
30	9.80	88.53
100	10.94	87.19
300	11.64	86.12
1000	10.69	87.09
k	$\overline{\text{ACC}}$	FGT
16	9.20	89.22
64	8.61	89.54

over the strongest prompt baseline (CODA-Prompt) $+0.72$ and -0.87 . The forgetting reduction is statistically significant on both benchmarks (Welch’s t -test: $p < 0.001$ on ImageNet-R and $p = 0.044$ on CIFAR-100); the accuracy improvement is significant on ImageNet-R ($p = 0.005$) and a clear mean advantage on CIFAR-100 ($p = 0.110$), where the high variance of the frozen-backbone, no-replay regime limits single-seed significance (Sec. 5.4). We further tuned EWC’s regularization strength on CIFAR-100 by grid search ($\lambda \in \{1, 10, 30, 100, 300, 1000\}$); its optimum $\lambda = 30$ reaches 9.58 ± 0.78 , still 0.49 points below FOLoRA, confirming that the advantage is not an artifact of an under-tuned baseline. All methods share a fixed frozen backbone and a single classifier head; the parameter count of FOLoRA, Seq-LoRA, EWC-LoRA, L2P and CODA-Prompt stays fixed across tasks, whereas O-LoRA allocates a new adapter per task and thus grows linearly with the number of tasks.

5.3. Ablation study

We ablate the two hyperparameters of FOLoRA — the regularization strength λ and the number of protected directions k — on Split CIFAR-100 (seed 0). Table 2 reports average accuracy and forgetting.

Accuracy is highest at $\lambda = 300$ and degrades on both sides, confirming that a moderately strong regularization best balances stability and plasticity: too small a λ under-protects the important directions, while too large a λ over-constrains the adapter and hurts plasticity. Increasing k from 16 to 64 does not help and slightly hurts, indicating that a compact protected

subspace ($k = 16$) already captures the dominant forgetting directions.

5.4. Limitations and discussion

The forgetting reduction of FOLoRA is statistically significant on both benchmarks. On CIFAR-100 the accuracy improvement is consistent in mean but noisy across seeds: the average accuracy of FOLoRA spans 0.086–0.127 over the ten seeds, whereas the baselines show a tighter spread. We attribute this variance to two sources. First, the importance kernel is estimated from a finite per-sample budget (up to 300 gradients per task), and its top- k approximation is sensitive to the particular samples that land in the tails of the estimate. Second, under a frozen backbone, a fixed-rank ($r=16$) adapter and no replay, the class ordering exerts a strong effect on the final subspace, so individual seeds drift apart. We further found that the epoch budget interacts with the method family: adapter-based Fisher methods degrade under longer training because over-fitting drives the per-sample gradients toward zero and thereby degenerates the estimated Fisher, whereas prompt-based methods benefit from longer training of their prompt keys. This motivates the per-family epoch budgets in Sec. 5.1. Even in this unfavorable regime FOLoRA never falls below the strongest baseline on either benchmark, which supports its robustness. A natural direction for future work is a stochastic (Monte-Carlo) Fisher estimate and class-order averaging to tighten the variance.

6. Conclusion

We proposed FOLoRA, a parameter- and memory-efficient method for continual learning that mitigates catastrophic forgetting in low-rank adaptation. FOLoRA maintains an accumulated second-order importance kernel for each adapted layer, forms an importance-weighted protected subspace from its top eigendirections, and regularizes the adapter to remain orthogonal to those directions in proportion to their importance. We derived a forgetting upper bound that decomposes into Fisher-weighted subspace overlaps and showed that FOLoRA’s regularizer minimizes its dominant term, thereby providing a principled stability–plasticity trade-off and subsuming equal-weight orthogonalization as a special case. Experiments on standard class-incremental benchmarks demonstrate consistent improvements over sequential LoRA, EWC-LoRA, O-LoRA, L2P and CODA-Prompt in both average accuracy and forgetting, with no replay buffer and no growth in the parameter count.

A natural direction for future work is to extend FOLoRA to task-agnostic settings with distribution shift (domain-incremental learning) and to language models, where the low-rank subspace geometry of different task families can be exploited more explicitly.

CRedit authorship contribution statement

Jiang Tao: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft.

References

- [1] R. M. French, Catastrophic forgetting in connectionist networks, *Trends in Cognitive Sciences* 3 (4) (1999) 128–135.
- [2] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, C. H. Lampert, icarl: Incremental classifier and representation learning, in: *CVPR*, 2017.
- [3] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, et al., Overcoming catastrophic forgetting in neural networks, *Proceedings of the National Academy of Sciences* 114 (13) (2017) 3521–3526.
- [4] F. Zenke, B. Poole, S. Ganguli, Continual learning through synaptic intelligence, in: *ICML*, 2017.
- [5] A. A. Rusu, N. C. Rabinowitz, G. Desjardins, et al., Progressive neural networks, in: *arXiv preprint arXiv:1606.04671*, 2016.
- [6] Z. Wang, Z. Zhang, C.-Y. Lee, et al., Learning to prompt for continual learning, in: *CVPR*, 2022.
- [7] J. S. Smith, L. Karlinsky, V. Gutta, et al., Coda-prompt: Continual decomposed attention-based prompting for rehearsal-free continual learning, in: *CVPR*, 2023.
- [8] X. Wang, T. Chen, Q. Ge, et al., Orthogonal subspace learning for language model continual learning, in: *Findings of EMNLP*, 2024.
- [9] E. J. Hu, Y. Shen, P. Wallis, et al., Lora: Low-rank adaptation of large language models, *ICLR* (2022).

- [10] Y.-S. Liang, W.-J. Li, Inflora: Interference-free low-rank adaptation for continual learning, CVPR (2024).
- [11] R. Aljundi, F. Babiloni, M. Elhoseiny, et al., Memory aware synapses: Learning what (not) to forget, in: ECCV, 2018.
- [12] A. Mallya, S. Lazebnik, Packnet: Adding multiple tasks to a single network by iterative pruning, CVPR (2018).
- [13] G. Saha, I. Garg, K. Roy, Gradient projection memory for continual learning, in: ICLR, 2021.
- [14] J. Martens, R. Grosse, Optimizing neural networks with kronecker-factored approximate curvature, ICML (2015).